

UserManager.sh — Script Explanation & Usage Guide

Comprehensive Technical Documentation & Terminal Command Reference

Script Name:	UserManager.sh	Interpreter:	Bash (/bin/bash)
Target System:	Linux (Ubuntu / Debian / RHEL)	Permissions:	Root / Sudo Required

1. Full Script Source Code

```
#!/bin/bash

# script to run without sudo
if [ "$EUID" -ne 0 ]; then
    exec sudo "$0" "$@"
fi

# $1 is the command (addGroup, addUser, delUser, delGroup, etc.)
# $2 is the first argument (name of group or user)
# $3 is the second argument (group name or path)

case "$1" in
    addGroup)
        groupadd "$2"
        echo "group $2 created"
        ;;
    addUser)
        USER_NAME="$2"
        GROUP_NAME="$3"

        #create user and assign the group
        useradd -m -g "$GROUP_NAME" "$USER_NAME"

        #set home directories permissions
        chmod 751 "/home/$USER_NAME"
        chown "$USER_NAME:$GROUP_NAME" "/home/$USER_NAME"

        #create sub directories team and ninja
        mkdir "/home/$USER_NAME/team"
        mkdir "/home/$USER_NAME/ninja"

        #set team dir permission
        chmod 770 "/home/$USER_NAME/team"
        chown "$USER_NAME:$GROUP_NAME" "/home/$USER_NAME/team"

        #set ninja dir permission
        chmod 777 "/home/$USER_NAME/ninja"
        chown "$USER_NAME:$GROUP_NAME" "/home/$USER_NAME/ninja"

        echo "user $USER_NAME is added to group $GROUP_NAME"
        ;;
    changeShell)
        chsh -s "$3" "$2"
        echo "shell updated for $2."
        ;;
esac
```

```

changePasswd)
    passwd "$2"
    echo "password updated for $2"
    ;;

ls)
    if [ "$2" == "User" ]; then
        cat /etc/passwd | cut -d: -f1
    elif [ "$2" == "Group" ]; then
        cat /etc/group | cut -d: -f1
    fi
    ;;

delGroup)
    groupdel "$2"
    echo "$2 group deleted"
    ;;

delUser)
    userdel -r "$2"
    echo "$2 user deleted"
    ;;

*)
    echo "Invalid command!"
    echo "Usage: $0 {addGroup|addUser|delGroup|delUser|changePasswd|changeShell|ls}"
    ;;

esac

```

2. Step-by-Step Technical Breakdown & Command Usage

Step 1: Automatic Sudo Privilege Escalation

The script starts by checking if it is executed with administrative (root) privileges. If not, it automatically prompts for the user's sudo password and elevates execution.

- `$EUID` : Environment variable storing the Effective User ID. Root has an EUID of `0`.
- `exec sudo "$0" "$@"` : Replaces the current process with `sudo`, passing the script name (`$0`) and all arguments (`"$@"`).

Step 2: Subcommand Dispatcher (`case "$1" in`)

Matches the first command-line parameter (`$1`) against supported operation flags.

Step 3: Creating a Group (`addGroup`)

Executes `groupadd "$2"` to create a new group on the Linux system.

```
$ ./UserManager.sh addGroup g1
```

```
[sudo] password for rahul:
group g1 created
```

Step 4: Creating a User with Provisioned Directories (`addUser`)

Creates user `$2` assigned to primary group `$3`, configures home directory ownership/permissions, and sets up mandatory subdirectories (`team` and `ninja`).

Target Directory	Permissions	Octal	Owner : Group	Access Description
<code>/home/USER_NAME</code>	<code>rwxr-x--x</code>	751	<code>USER:GROUP</code>	Owner full access; Group read/execute; Others traverse only.
<code>/home/USER_NAME/team</code>	<code>rwrxwx---</code>	770	<code>USER:GROUP</code>	Owner & Group full read/write/execute; Others no access.
<code>/home/USER_NAME/ninja</code>	<code>rwrxwxrwx</code>	777	<code>USER:GROUP</code>	World readable, writable, and executable by any system user.

```
$ ./UserManager.sh addUser u1 g1
```

```
user u1 is added to group g1
```

Step 5: Changing User Password (`changePasswd`)

Invokes Linux's interactive `passwd` utility to set/reset user credentials.

```
$ ./UserManager.sh changePasswd u1
```

```
New password:
Retype new password:
passwd: password updated successfully
password updated for u1
```

Step 6: Updating User Default Shell (`changeShell`)

Uses `chsh -s` to change a user's default login shell stored in `/etc/passwd`.

```
$ ./UserManager.sh changeShell u1 /bin/zsh
```

```
shell updated for u1.
```

Step 7: Listing System Users and Groups (`ls`)

Parses system database files (`/etc/passwd` and `/etc/group`) using `cut` with a colon `:` delimiter to extract account names.

```
$ ./UserManager.sh ls User
```

```
root
daemon
rahul
u1
```

```
$ ./UserManager.sh ls Group
```

```
root
sudo
rahul
g1
```

Step 8: Deleting Users and Groups (`delUser` & `delGroup`)

Removes user accounts (along with home directory cleanup via `-r`) or system groups.

```
$ ./UserManager.sh delUser u1
```

```
u1 user deleted
```

```
$ ./UserManager.sh delGroup g1
```

```
g1 group deleted
```

3. Verification & Auditing Commands

To verify that directory permissions and active shell paths were configured accurately by the script, use the following terminal commands:

1. Check Directory Tree Structure:

```
$ sudo tree /home/u1
```

```
/home/u1
├─ ninja
└─ team
```

2. Inspect Detailed Permissions and Ownership:

```
$ sudo ls -la /home/u1
```

```
drwxr-x--x 4 u1 g1 4096 Jul 30 12:30 .
drwxr-xr-x 3 root root 4096 Jul 30 12:30 ..
drwxrwxrwx 2 u1 g1 4096 Jul 30 12:30 ninja
drwxrwx--- 2 u1 g1 4096 Jul 30 12:30 team
```

3. Verify Configured User Shell:

```
$ getent passwd u1 | cut -d: -f7
```

```
/bin/zsh
```